



**Northumbria
University**
NEWCASTLE

Online Videos Week 3

Topic 1: Axial Stress & Truss Elements

KB5034 Mechanics and Finite Element Analysis

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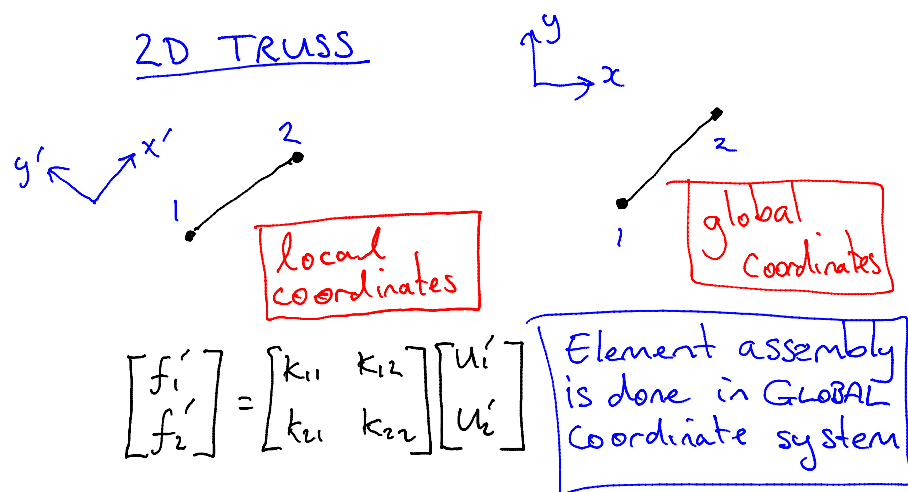
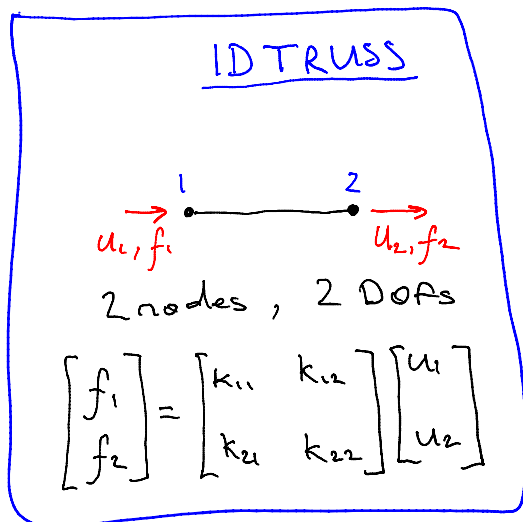
Department of Mechanical & Construction Engineering

Learning Outcome 1.4

By the end of this lesson, students will be able to...

Assemble the stiffness matrix for simple 2D truss structures with appropriate boundary conditions.

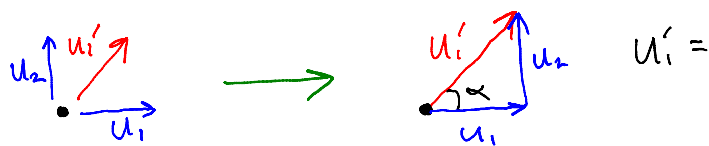




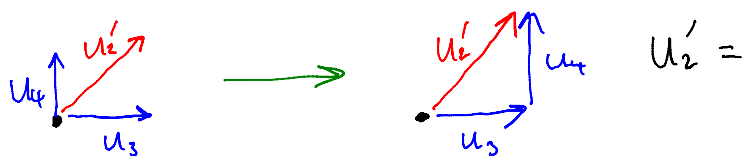
for 2D truss elements, we must transform local coordinates to the global coordinate system.

Displacement Transformation

At node 1:



At node 2:



In matrix form:

$$\begin{bmatrix} u'_1 \\ u'_2 \end{bmatrix} = \underbrace{\begin{bmatrix} & \\ & \end{bmatrix}}_{\text{Transformation matrix}} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}$$

Local Global

$$\underline{u'_e} = \underline{T} \underline{u_e} \quad \text{element}$$

Force Transformation

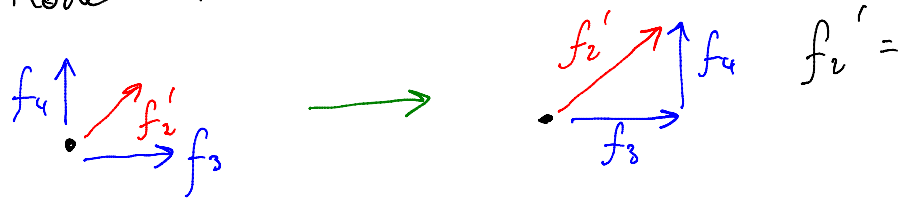
At node 1:



At node 1:



At node 2:



In matrix form:

$$\begin{bmatrix} f_1' \\ f_2' \end{bmatrix} = \begin{bmatrix} \end{bmatrix}$$

$$\begin{bmatrix} f_1' \\ f_2' \\ f_3' \\ f_4' \end{bmatrix}$$

$$\underline{f_e'} = \underline{T} \underline{f_e}$$

Transformation to Global System

Local

Global

$$\underline{u_e'} = \underline{T} \underline{u_e}$$

$$\underline{f_e'} = \underline{k_e'} \underline{u_e'}$$



$$\underline{f_e} = \underline{k_e} \underline{u_e}$$

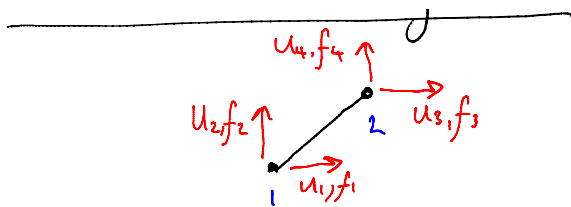
$$\underline{f_e'} = \underline{T} \underline{f_e}$$

Substitute u_e' and f_e' into local equation:

Move to RHS

Putting this all together





$$\underline{f}_e = \underline{I}^T \underline{K}_e' \underline{I} \underline{U}_e$$

$$\begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix} = \begin{bmatrix} \cos \alpha & 0 \\ \sin \alpha & 0 \\ 0 & \cos \alpha \\ 0 & \sin \alpha \end{bmatrix} \begin{bmatrix} \frac{EA}{l} & -\frac{EA}{l} \\ -\frac{EA}{l} & \frac{EA}{l} \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha & 0 & 0 \\ 0 & 0 & \cos \alpha & \sin \alpha \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}$$

Dot product [order is important!]

$$\begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix} = \frac{EA}{l} \begin{bmatrix} \cos \alpha & -\cos \alpha \\ \sin \alpha & -\sin \alpha \\ -\cos \alpha & \cos \alpha \\ -\sin \alpha & \sin \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha & 0 & 0 \\ 0 & 0 & \cos \alpha & \sin \alpha \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}$$

$$\begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix} = \frac{EA}{l} \begin{bmatrix} \cos^2 \alpha & \cos \alpha \sin \alpha & -\cos^2 \alpha & -\cos \alpha \sin \alpha \\ \cos \alpha \sin \alpha & \sin^2 \alpha & -\cos \alpha \sin \alpha & -\sin^2 \alpha \\ -\cos^2 \alpha & -\cos \alpha \sin \alpha & \cos^2 \alpha & \cos \alpha \sin \alpha \\ -\cos \alpha \sin \alpha & -\sin^2 \alpha & \cos \alpha \sin \alpha & \sin^2 \alpha \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix}$$

$\underline{K}_e = \underline{I}^T \underline{K}_e' \underline{I}$

Summary of Finite Element Method

This applies to all static problems with any element type.

Step 1

Step 2

Step 3

Step 4

Step 5